

Donnée du travail de bachelor

Vincent Cordola

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Study programme	Academic year	Thesis reference
ISC	2025-26	ISC-ID-26-6
Principal	Expert	Place of execution
☐ HES—SO Valais-Wallis ☐ Industry ☒ McGill Montreal, Canada	Dr. Nikhil Bhagwat 3801 University Street Montreal, Canada nikhil.bhagwat@mcgill.ca	☐ HES—SO Valais-Wallis ☐ Industry ☒ MOVE, McGill Montreal, Canada
Confidential work	Professor	Co-supervisor
☐ yes ☒ no	Prof. Oscar Esteban	Prof. Jean-Baptiste Poline
Date version	May 12, 2026 - v1.30	

Title	Peer-to-Peer (P2P) Dataset Federation
Description	This TB will deliver a proof-of-concept demonstrating that a Jupyter notebook can open and query a federated collection of Parquet files distributed across an ad-hoc peer-to-peer (P2P) network of 2–5 machines, with access that appears local and seamless to the user. The P2P transport layer is provided by iroh, an existing Rust-based networking toolkit; the student’s implementation work focuses on the Python client layer that discovers, fetches, caches, and exposes distributed Parquet files to standard dataframe tools (PyArrow, pandas, DuckDB). The motivating use case is the sharing of Image Quality Metrics produced by MRIQC, a neuroimaging quality control tool, across institutions without centralized infrastructure. The project prioritizes a working end-to-end demonstration over low-level protocol development.
Objectives	Knowledge, skills, and competences to be acquired: 1. Understand the fundamentals of peer-to-peer networking (peer discovery, data integrity, NAT traversal) and configure an ad-hoc iroh network using existing iroh tooling. 2. Design and implement a Python client library that discovers available Parquet files on the network, fetches missing files into a local cache, and presents the distributed collection as a single dataset accessible from a Jupyter notebook. Validate application logic on a containerized testbed (Docker Compose). 3. Deliver a reproducible demonstration package: a working Jupyter notebook, setup scripts, dataset conventions, and a short evaluation report documenting performance, limitations, and recommended next steps. 4. <i>Bonus:</i> Validate the prototype on 2–5 physical machines under realistic network conditions (peer join/leave, partial availability, NAT traversal).

Signature		Deadlines
Head of programme orientation	Student	Theme attribution : March 03, 2026 Start of bachelor thesis : May 11, 2026 Final report submission : August 12, 2026 12:00 Oral defence : Semaines du 17 et 25 août 2026 Poster exhibitions : August 28, 2026 - HEI Pitch challenge : August 31, 2026 - Monthey
		

Addendum — Bachelor thesis description

Project type	☐ Exploratory ☒ Implementation
Data dependencies	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 1 No data 2 Data collected, cleaned and ready to be used 3 Data collected, cleaning needed 4 Data not collected but available online 5 Data not collected and dependent on external actors
Material dependencies	☒ 1 ☐ 2 ☐ 3 1 No specific material needed 2 Material is readily accessible on-site 3 Material is not readily accessible and needs to be acquired
Extra information	Constraints and risks. The primary constraint is the dependency on iroh, a relatively young Rust ecosystem whose APIs may evolve during the project; this is mitigated by pinning dependency versions early and limiting the prototype to a small, well-documented subset of iroh primitives (blobs for transfer, optionally gossip for discovery). A second constraint is the need for a multi-machine testbed (2–5 devices) with heterogeneous network conditions. On the student side, the main risk is the learning curve of combining Rust systems programming with Python integration in a single project; this is contained by keeping the Rust component minimal (a thin node agent) and relying on well-established Python tooling (PyArrow, pandas, DuckDB) for the user-facing layer. Finally, because iroh-blobs’ verified range requests have not yet been tested against Parquet’s footer-first read pattern, the Parquet partial-read optimization is treated as optional: the baseline design downloads full files into a local cache, which is sufficient for the dataset sizes expected in this proof of concept. Proposed workplan (11/05/2026–17/08/2026): - Onboarding and technical design (Weeks 1–2 11/05–24/05) - Rust & iroh 101 crash-course (starting before TB). - Define the dataset layout conventions. - Core P2P sharing prototype (Weeks 3–6 25/05–21/06) - Rust MVP: (i) join an ad-hoc iroh network, (ii) advertise Parquet files, and (iii) fetch Parquet files from peers. - Validation on a Docker Compose network of 3–5 containers. - Python/Jupyter integration and connectivity validation (Weeks 7–10 22/06–19/07) - Python client layer: a unified Jupyter interface ensuring required files are present locally via on-demand retrieval. - Migrate the testbed to virtual machines with separate network stacks to exercise NAT traversal and relay fallback for the first time. - Produce a first working notebook that discovers the network dataset, loads it into standard dataframe tooling, and runs queries spanning multiple peers. - Robustness and evaluation in a 2–5 machine testbed (Weeks 11–13 20/07–09/08) - Test behavior under realistic conditions: peers joining/leaving, partial availability, varying bandwidth. - Improve reliability (resume partial transfers, cache behavior, error handling) and document known limitations. - Optional (if feasible): evaluate byte-range or partial reads to reduce transfers for larger Parquet artifacts. - Packaging, documentation, and final demonstration (Week 14 10/08–17/08) - Finalize the notebook as the primary demonstration artifact. - Produce setup scripts/instructions for reproducing the demo on 2–5 machines. - Write a short evaluation report summarizing results, constraints, and recommended next steps for downstream integration work.